HARIS WAHEED BHATTI

BIM Specialist | Information Manager (ISO 19650)

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Date of Birth: 25.12.2001 | Nationality: Pakistani

Open to relocating to Italy; available immediately.

PROFILE

Civil engineer with a European Master's in Building Information Modelling and a strong software background: Python, TypeScript, agentic AI systems, and parametric design in Dynamo. Built an LLM agent for structured IFC data querying and an open-source IFC-to-Neo4j converter, and delivered parametric façade optimisation and federated BIM coordination workflows.

PROJECTS

Federated BIM Model – Pakistani-Themed Expo, Guimarães | IFC, Autodesk Construction Cloud 10.2025

- Owned the federated model and clash-detection workflow across architecture and MEP disciplines, validated against Employer's Information Requirements under ISO 19650.

ifctoneo4j – Python package (PyPI) | Python, IfcOpenShell, Neo4j, Cypher 2026

- Open-source library converting IFC models (IFC2X3, IFC4, IFC4X3) into Neo4j property graphs aligned to Linked Building Data ontologies, with no RDF intermediate. Published to PyPI: `pip install ifctoneo4j`.

BIMGraphExplorer – LLM agent for IFC question answering | Python, MCP, Neo4j 2026

- Adaptive querying system exposing four read-only tools over the Model Context Protocol; profiles the graph schema and issues iterative Cypher queries conditioned on prior results.
- Raised answer accuracy by 33 percentage points and source transparency from 12.5% to 99.7% on a fixed model and dataset, through prompt design alone.
- Benchmarked against a published direct-IFC baseline under matched conditions (same model, same validated judge; 23 IFC datasets, 367 questions, 15 projects). Overall accuracy 72.0% against the baseline's 82.3%, but reached source-transparency parity (99.7% versus 100%) and outperformed it on 5 of 23 datasets, by up to 24 points on one.

EDUCATION

MSc, European Master in BIM (BIM A+) | UL FGG, Ljubljana, Slovenia ...... 10.2025 – 07.2026

- Field of study: Information Management in Construction. Completed the accelerated thesis track, finishing full MSc requirements in 10 months. Final grade 9.27/10.
- Thesis: *Graph-Mediated LLM Exploration for IFC Question Answering: A Benchmark Comparison Against Direct-IFC Approaches*. Supervisor: Prof. Žiga Turk. hdl.handle.net/20.500.12556/RUL-185169
- Applied coursework spanning the BIM delivery chain: IFC schema auditing and model remediation, parametric façade optimisation in Dynamo, process modelling and workflow improvement, 5D quantity take-off and cost estimation, and scan-to-BIM from point-cloud capture.

BE Civil Engineering | NUST, Islamabad, Pakistan. Final grade 3.25/4.00 ...... 09.2020 – 06.2024

EXPERIENCE

Civil Engineering Intern | Fatima Fertilizer Company, Pakistan 07.2023

- Modelled a 6,804 m^2 rock phosphate storage shed in 3D from 2D drawings, coordinating with stakeholders on design refinement.

Contract Management Intern | Engineering and Management Services, Pakistan 06.2023

- Reviewed contract reports and arbitration documentation under FIDIC conditions; presented findings to the internal legal team.

Research Intern (remote) | Southern Illinois University Edwardsville, USA 07.2023

- Synthesised academic literature to support a faculty-mentored research proposal.

SKILLS

Programming & Data: Python, TypeScript, Neo4j, Cypher, knowledge graphs, Git, ArcGIS
AI / LLM: Agentic systems, Model Context Protocol (MCP), RAG, LLM evaluation and benchmarking
BIM & AEC: Revit, Dynamo (parametric design), clash detection, federated models, IFC (IFC2X3, IFC4, IFC4X3), ISO 19650, Employer's Information Requirements, Autodesk Construction Cloud, IfcOpenShell, Linked Building Data ontologies
Languages: English (C1)